Bootstrapping Vision-Language-Action Models for New Robot Embodiments with Staged Reinforcement Learning

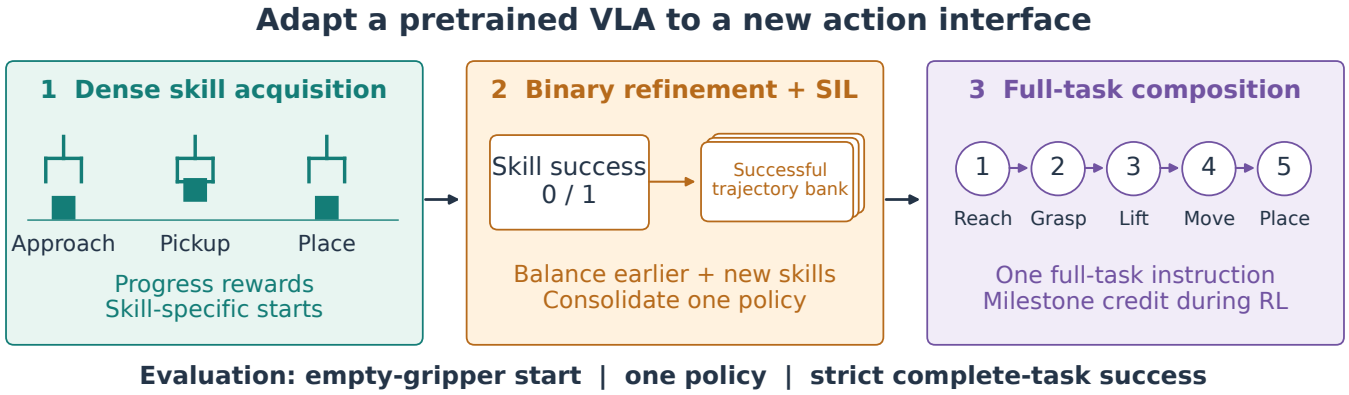


Fig. 1. Proposed pipeline. Dense feedback acquires skills; binary refinement and balanced self-imitation consolidate them; milestone-based RL learns the complete task under one instruction. Skill-specific resets are used during acquisition. Full-task evaluation starts with an empty gripper and uses no external switching between specialists.

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***Abstract*—Adapting a vision-language-action (VLA) model to a new robot is difficult when its action interface produces few successful rollouts. We propose a three-stage reinforcement learning pipeline: dense skill acquisition, binary refinement with self-imitation, and milestone-based full-task composition. Dense rewards guide early exploration. A balanced bank of successful trajectories helps recover earlier skills, and milestone credit trains one policy to execute the complete instruction. We implement the pipeline with pretrained SmolVLA, LoRA, and an embodiment-specific residual controller. The controller achieves 61.43% strict pick-and-place success on CDPR simulation development scenes. A retention experiment increases reaching success from 8.0% to 31.1% while preserving placement performance. On LIBERO-Spatial, phase-aware dense feedback reaches 65.3% success versus 61.7% with terminal-only reward. These results support structured feedback and skill consolidation; controlled ablations of the complete schedule remain open.**

***Index Terms*—Robot learning, reinforcement learning, manipulation, vision-language-action models, robot adaptation.**

I. Introduction

Vision-language-action (VLA) models provide useful visual and language representations for robot control. A new embodiment can still require a different action mapping. Before this mapping is learned, complete-task success may be too rare to provide a useful binary learning signal.

Consider "put the apple on the plate." The robot must approach, grasp, lift, transport, and release the object. Terminal failure alone does not distinguish a missed grasp from a failed release. Learning these skills separately improves the starting point, but training a new skill can also erase earlier behavior. The final policy must both retain the skills and connect them under one instruction.

We address these problems with the pipeline in Fig. 1. Dense rewards first guide individual skill learning. Binary rewards then target skill completion, and balanced self-imitation consolidates successful experience across instructions. Finally, milestone-based RL trains complete-task execution from ordinary empty-gripper starts.

We study embodiment adaptation with SmolVLA on a cable-driven parallel robot (CDPR). A learned residual controller adapts a retained pretrained action expert to the target interface. We also study reward construction with OpenVLA-OFT on LIBERO, where an established action policy lets us examine intermediate feedback separately from learning a new interface.

Our contributions are (i) a staged skill-learning and composition pipeline, (ii) a residual implementation with explicit milestone credit, and (iii) empirical evidence on skill recovery, full-task execution, and phase-aware feedback. LIBERO evaluates reward design; it does not evaluate the complete adaptation schedule.

II. Related Work

VLA adaptation. OpenVLA [1], SmolVLA [2], and OpenVLA-OFT [3] provide pretrained policies and adaptation methods. We retain SmolVLA's pretrained expert and learn a separate residual. Our LIBERO implementation instead updates discrete action-token likelihoods in an OpenVLA-OFT-based policy.

Residual learning and consolidation. Probe, Learn, Distill combines residual RL, autonomous data collection, and distillation [4]. These ingredients are established; our focus is their use in a schedule that links skill acquisition, retention, and composition. Here self-imitation means supervised fitting of

successful policy-generated experience, including experience retained from earlier policies.

Intermediate feedback and RL. SimpleVLA-RL applies group-relative optimization to VLA rollouts [5], following GRPO [6]. Binary group rewards provide no contrast when all candidates have the same outcome [7], [8]. Intermediate progress can create contrast earlier. Our shaping rule is task-specific and does not claim the policy-invariance guarantees of potential-based shaping [9]. We also distinguish latent samples from clipped executed actions when computing likelihoods [10].

III. Problem Formulation

A policy receives camera images, proprioception, and instruction ℓ, and predicts an action chunk. The controller executes a prefix before replanning. Let h_t denote the available observation history and c the scene initialization. We optimize binary complete-task success $Y(\tau)$:

$$J(\theta) = \mathbb{E}_{(c,\ell)\sim\rho_{\text{full}},\, \tau\sim P_\theta(\cdot|c,\ell)}[Y(\tau)]. \tag{1}$$

Individual skills have their own instructions, success predicates, and reset distributions ρ_k. Placement from a held-object start is therefore a skill test, not full pick-and-place. A training stage is a step in the learning schedule; a task phase is progress within an episode.

The CDPR action has five channels: Cartesian translation, yaw, and gripper control. Strict success requires grasp, held lift, intentional release, and native placement, without carry-slip or wrong-placement events. LIBERO uses its native goal verifier [11]. We report these different evaluation settings separately.

IV. Method

A. Dense Acquisition, Binary Refinement, and Self-Imitation

We first train individual skills with dense rewards for approach, pickup, and transport/release. Distance reduction and intermediate manipulation events provide feedback before the policy can reliably complete a skill. We then use binary skill-success rewards to refine completion.

Successful policy trajectories form instruction-specific banks $\mathcal{D}_k^+$. Self-imitation fits their executed actions with a balanced mixture:

$$\mathcal{L}_{\text{SI}} = \sum_k \alpha_k\, \mathbb{E}_{(h,\ell,a)\sim\mathcal{D}_k^+}\left[\|\mu_\theta(h,\ell) - a\|^2\right]. \tag{2}$$

We balance valid decision rows across instructions and mask unused chunk slots and terminal tails. Stored images and executed actions are retained; policy-dependent priors and visual features are refreshed before fitting. Historical scripted or teacher-assisted data are identified separately from policy self-imitation. The exact acquisition budgets and dense-to-binary switches remain **[To complete: training manifest]**.

B. SmolVLA with an Embodiment-Specific Residual

Figure 2 shows the action path. We retain the pretrained SmolVLA expert and map its output to a target prior $p \in \mathbb{R}^{8\times5}$. A frozen 512-dimensional visual projection and six proprioceptive values form the 518-dimensional observation vector x. The residual also receives the flattened 40-dimensional prior:

$$\mu_\theta(x,p) = \tanh(p + s_r \tanh f_\theta([x,p])). \tag{3}$$

The MLP has widths $558 \to 1024 \to 1024 \to 40$, with $s_r = 1$. It predicts eight action slots; four are executed before replanning. LoRA adapts the action expert in earlier phases. The evaluated full-task continuation freezes the inherited VLA and LoRA while training the residual and exploration parameters. These dimensions follow the inspected code and configuration; the historical resolved run configuration remains to be archived.

C. Milestone-Based Full-Task Composition

One policy learns the complete instruction from empty-gripper starts. It receives no external sequence of skill prompts at evaluation. Three binary milestones M_{ik} mark approach, held pickup, and strict placement, with $M_{i3} = Y_i$. Within each scene group, milestone returns are centered and normalized to give advantages A_{ik} and usability masks U_{ik}. Constant-return groups give no milestone contrast.

For approach and pickup, we add terminal credit while suppressing a negative terminal bonus on an achieved milestone:

$$\widetilde{A}_{ik} = U_{ik}A_{ik} + \beta U_{i3} T_\kappa(A_{i3}, M_{ik}), \tag{4}$$

$$T_\kappa(a,m) = \begin{cases} \kappa a, & m = 1,\ a < 0, \\ a, & \text{otherwise.} \end{cases} \tag{5}$$

We use $\beta = 1$ and $\kappa = 0$. Placement uses its terminal advantage once. Figure 3 illustrates the rule. Nonnegative stage weights, a clipped PPO-style surrogate, entropy and action penalties determine the update. Prefix protection applies to this credit component; shared neural parameters can still interfere across stages.

Exploration samples $u_t = \mu_\theta + b_t + \sigma_\theta \odot \varepsilon_t$ and executes $a_t = \text{clip}(u_t, -1, 1)$. Likelihood ratios score the recorded unclipped sample under the Gaussian conditioned on the realized episode offset b_t. Scoring clipped actions as Gaussian samples would misrepresent mass at the action bounds [10].

D. Phase-Aware Dense Feedback

For LIBERO, a simulator-state tracker follows reach, grasp, lift, move, and place. It rewards only new progress beyond the best value B_t reached in the active phase, $\Delta p_t^+ = \max(p_t - B_t, 0)$. The baseline resets at phase advancement. With c_t newly completed phases and terminal verifier output y_t,

$$r_t = \lambda y_t + \text{clip}(w_p \Delta p_t^+ + w_c c_t, -C, C). \tag{6}$$

Setting $w_p = 0$ gives phase-boundary feedback; $w_p > 0$ also rewards within-phase progress (Fig. 4). The terminal reward remains unchanged outside the shaping clip. Exact phase predicates and progress functions remain **[To complete: reward implementation]**.

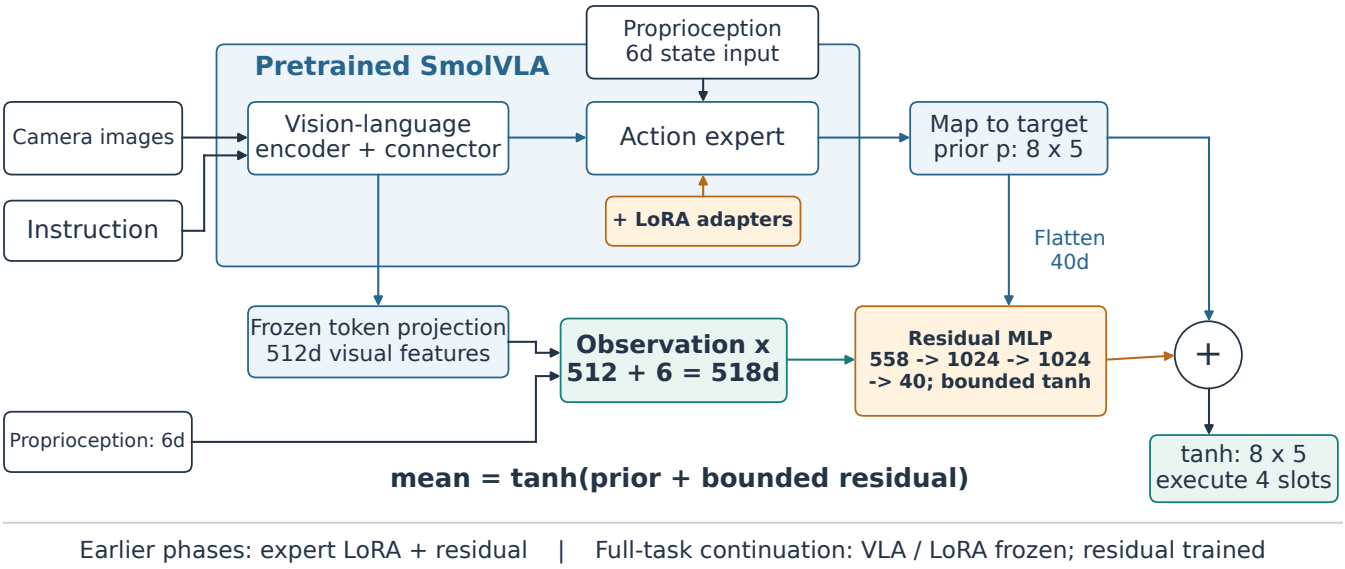


Fig. 2. SmolVLA adaptation. The pretrained expert supplies a mapped action prior. The added residual receives 518 observation features plus the 40-value prior. LoRA modifies the expert in earlier phases; inherited VLA/LoRA weights are frozen during the evaluated full-task continuation. The residual feature branch uses a frozen projection. Dimensions refer to the CDPR implementation.

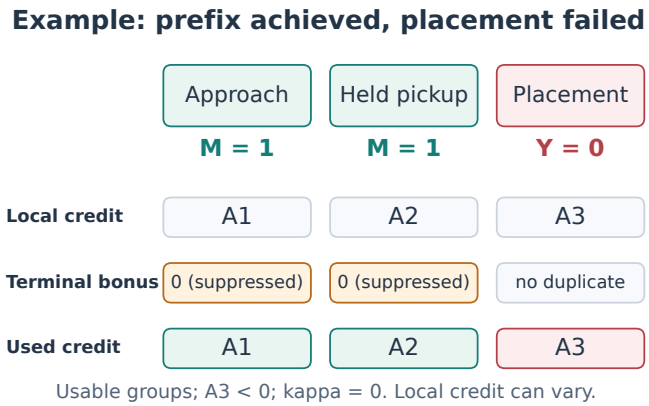


Fig. 3. Usable-group example: credit for a rollout that reaches and lifts the object but fails placement. Local milestone credit is retained. The negative terminal bonus is suppressed on achieved prefix stages; the placement stage retains its own terminal credit. Colors show credit routing, not measured gradient magnitudes.

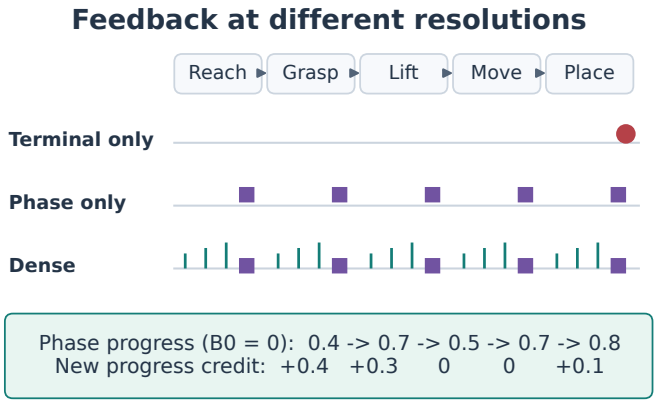


Fig. 4. Feedback resolution. Terminal-only reward distinguishes final success. Phase rewards also mark completed subgoals. Dense shaping credits new progress within a phase; revisiting an earlier maximum earns no further progress reward. This is a schematic, not a training curve.

Group filtering is separate: a group is retained when its terminal success fraction lies in $[L, U]$. A lower bound above zero can reject all-failure groups even when their dense returns differ.

V. Why Intermediate Feedback Helps

For G independent candidates from one scene, each with binary success probability p, the probability of a mixed-outcome group is

$$P_{\text{mixed}} = 1 - p^G - (1-p)^G. \tag{7}$$

At $G = 8$, it is 21.6% for $p = 0.03$ and 94.2% for $p = 0.30$. Milestones can therefore provide contrast before full success becomes common. Dense progress can distinguish candidates within one milestone level. Neither guarantees useful learning if the feedback is constant or the collector rejects the group.

With separate Bernoulli gates and raw on-policy ascent, suppressing negative completion credit on achieved gates makes each gate's update nonnegative. This finite-model property does not extend directly to a shared neural controller, PPO clipping, or optimizer momentum. We use it to explain the credit rule, without claiming monotonic improvement [12].

VI. Experimental Setup

A. SmolVLA on the CDPR

Simulation uses MuJoCo Warp/MJ-Lab, four object categories (apple, orange, potato, tomato), and plate/bowl destinations. Full-task evaluation covers 512 development scenes, excluding the 128-scene in-run panel, with four repeats per

scene and a 128-decision horizon. It uses the residual mean, VLA prior noise scale 1, no settling decisions, and no teacher or yaw assistance. Every attempt, including numerical failures, remains in the denominator. The controller comes from one training lineage; evaluation repeats are not independent training seeds.

The retention study uses the family-specific protocols in the campaign report. Each family retains its recorded reset protocol. Three-family consolidation uses 6,009 reaching, 6,001 plate, and 5,694 bowl decision rows. We report round-level observations separately from complete-task performance. The source report does not provide a complete episode-level statistical record.

B. OpenVLA-OFT on LIBERO

We use SimpleVLA-RL on ten LIBERO-Spatial tasks with two NVIDIA L40 GPUs, rollout batch 32, and LoRA rank/alpha 16. The implementation optimizes discrete action-token likelihoods. Aggregate experiments share a 61.0% initialization and use 300 validation rollouts per reported endpoint.

The supplied CSV specifies $w_p = w_c = C = 0.05$ for dense shaping. The primary dense and terminal-only conditions use $\lambda = 5$; the latter disables shaping. Filtered dense adds $[L, U] = [0.1, 0.9]$. The $\lambda = 1$ phase-only condition uses $w_p = 0, w_c = 0.20, C = 0.05$. Endpoint update counts differ across configurations and are reported in Table II.

The low-success study uses `task_1` and filtering $[0.1, 1.0]$. Its initialization is 2.0%, measured over 1,000 rollouts. Final evaluation denominators, independent seeds, wall time, and the canonical task name remain to be supplied.

C. Additional Robot and Hardware Slots

Figure 6 reserves illustrations for the five-DoF WX200 and its task sequence. Its action mapping, calibration, and evaluation protocol remain to be specified. Hardware success rates are left unfilled until the trial logs are available.

VII. Results

A. Complete-Task Execution

The adapted SmolVLA controller completes 1,258 of 2,048 attempts (61.43%) from empty-gripper starts. Table I reports its task outcomes. Held lift occurs in 85.89% of attempts, but only 71.52% of held-lift attempts reach strict completion. Transport and release remain a major bottleneck. Plate and bowl completion rates are 68.58% and 54.22%, respectively. These are development results from one training lineage.

B. Forgetting and Recovery with Balanced Self-Imitation

Figure 5 reports skill-level forgetting and recovery. In the forgetting measurements, reaching falls from a mean of 64.1% to 5.3% after placement RL, then to 1.8% after placement-only imitation. In a separate consolidation measurement, balanced self-imitation increases reaching from 8.0% to 31.1%. Plate and bowl placement change from 50.0% to 52.3% and from 23.05% to 27.65%. The panels use the reported family protocols and are not joined into a continuous learning curve.

TABLE I
CDPR complete-task development evaluation: 512 scenes, four repeats each. Rates use all 2,048 attempts. Event flags can overlap.

Outcome	Attempts (%)
Physical grasp	89.65
Held lift	85.89
Strict completion (1,258/2,048)	61.43
Carry slip	23.83
Wrong placement	19.63
Numerical failure	1.37

TABLE II
LIBERO reward comparison and low-success task. Success and Δ are percentages and percentage points. Aggregate endpoints use 300 validation rollouts; final task_1 denominators are unreported. Different update budgets are shown explicitly.

Configuration	Updates	Success	Δ
LIBERO-Spatial: initial 61.0%			
Terminal only, $\lambda = 5$	160	61.7	+0.7
Dense, $\lambda = 5$	160	65.3	+4.3
Filtered dense, $\lambda = 5$	50	64.7	+3.7
Dense, $\lambda = 1$	110	61.0	+0.0
Phase only, $\lambda = 1$	110	62.3	+1.3
task_1: initial 2.0% (1,000 rollouts)			
Terminal only	20	11.0	+9.0
Dense	20	18.0	+16.0

The bank includes reaching experience from a retained policy and the placement policy's own successes. This supports consolidation from stored successful experience. It does not establish a causal effect of SIL on final full-task success: later full-task imitation attempts also failed to improve performance. Matched schedule and retention ablations remain necessary.

C. Phase-Aware Feedback on LIBERO

At 160 updates, primary dense feedback reaches 65.3% success versus 61.7% for terminal-only reward, an observed difference of 3.6 percentage points (Table II). Filtered dense reaches 64.7% at 50 updates. Different budgets prevent a direct efficiency claim across all rows.

At $\lambda = 1$, phase-only feedback reaches 62.3% and dense shaping 61.0%. Their phase weights also differ, so this comparison does not isolate within-phase progress. On `task_1`, dense and terminal-only training reach 18% and 11% after 20 updates, from the 2% initialization. These descriptive results support intermediate feedback under the tested settings. They do not establish a statistically resolved advantage, a throughput gain, or the benefit of the complete adaptation schedule.

D. Reserved Robot Illustrations

The WX200 layout in Fig. 6 reserves the apparatus, control axes, and a time-ordered task sequence. Actual images and measurements will replace these slots. No hardware result is reported in this draft.

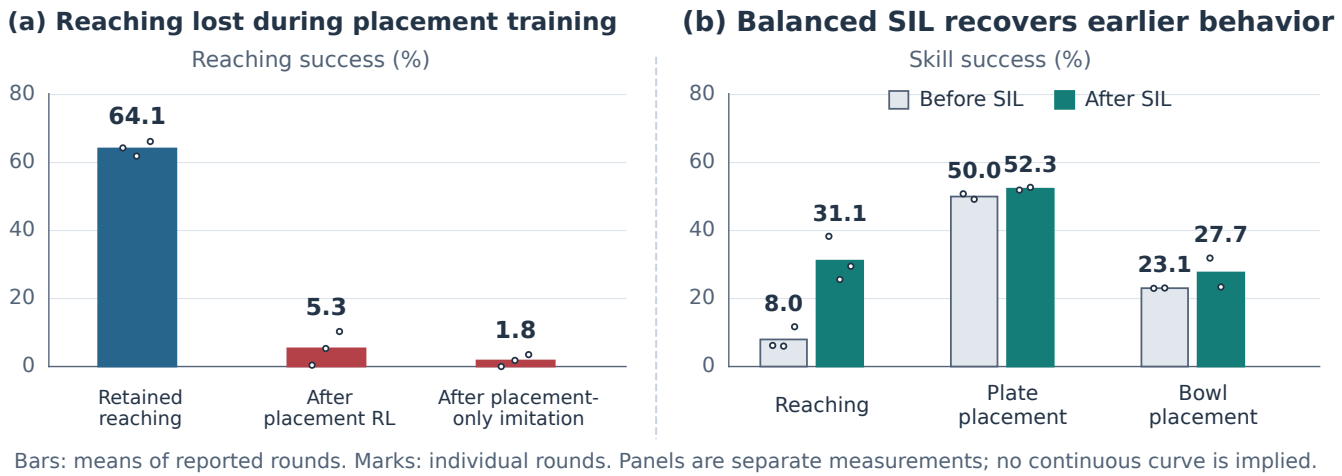


Fig. 5. Measured skill forgetting and recovery. Left: reaching after placement RL and placement-only imitation (three reported rounds per condition). Right: before and after balanced three-family self-imitation (three reaching rounds, two placement rounds). Small marks show reported rounds; bars show their means. The two panels describe separate measurements. Rounded source rates are used; raw evaluation denominators and paired episode records are unavailable.

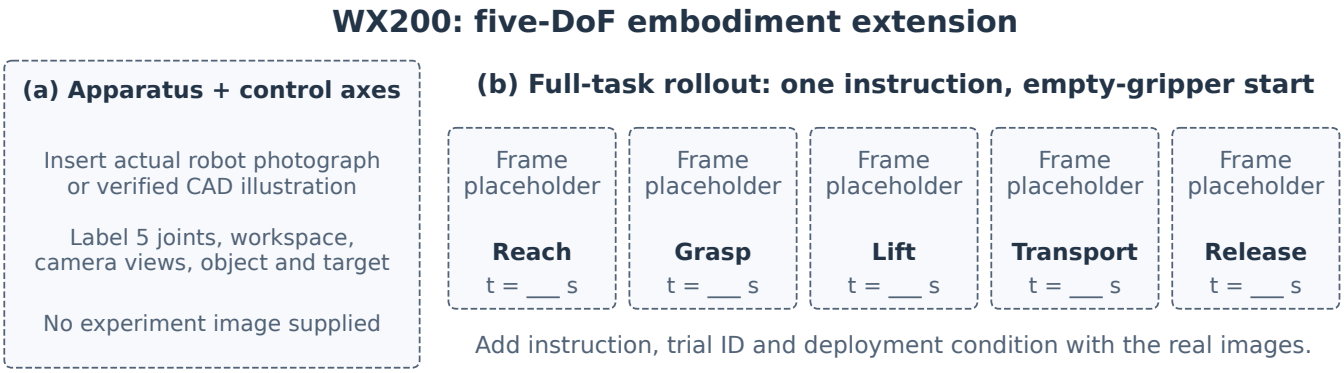


Fig. 6. Reserved WX200 illustrations. (a) Five-DoF robot, workspace, cameras, and control-axis annotation. (b) Actual rollout frames for reach, grasp, lift, transport, and release. Add timestamps and the task instruction. The current panels are layout placeholders and contain no experimental evidence.

VIII. Discussion and Limitations

The pipeline assigns a distinct role to each step: dense feedback guides discovery, binary rewards target completion, balanced imitation consolidates skills, and milestone credit supports composition. The retention results show why consolidation must include earlier instructions. They also show that recovery is partial.

The complete schedule is evaluated on one CDPR training lineage with a retained SmolVLA expert. LIBERO evaluates reward construction with a trained action interface. Neither study isolates every schedule component. Controlled dense-to-binary, SIL, and composition ablations, independent training seeds, and a locked test set are still needed. Manual rewards, inherited training, and historical teacher/scripted data remain part of the method's scope. WX200 and physical deployment results remain pending.

IX. Conclusion

We presented dense skill acquisition, binary refinement with self-imitation, and milestone-based full-task composition for VLA adaptation. A SmolVLA residual controller achieves 61.43% strict CDPR development success. Balanced consolidation partially recovers an earlier skill, and LIBERO experiments show an observed benefit of phase-aware feedback. Further controlled experiments are needed to establish the contribution of each training stage.

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